



## Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no. ....

Name of the Course: B. Tech (AIML and AIDS)

Semester: II

Name of the Paper: Fundamentals of Artificial Intelligence and Machine Learning

Paper Code: TCS203

Time: 1.5 hour

Maximum Marks: 50

**Note:**

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

**Q1. (10 Marks)**

- a. Define PEAS description and explain its components with an example (e.g., autonomous car or vacuum cleaner). CO1

OR

- b. Discuss the impact of Artificial Intelligence on society and industry. Discuss both positive and negative implications. CO1

**Q2. (10 Marks)**

- a. Explain the working of Depth-First Search (DFS) algorithm with the help of an example and analyse the algorithm based on the following: CO2

- i. Completeness
- ii. Optimality
- iii. Time Complexity

OR

- b. Consider the following state space graph consisting of five nodes:

Nodes: S (Start), A, B, C, G (Goal) CO2

Edges:

- S → A
- S → B
- A → C
- B → E
- E → D
- D → G

- i. Apply the Breadth-First Search (BFS) algorithm to find a path from S to G.
- ii. Identify the final path obtained.
- iii. Comment on whether BFS guarantees the optimal solution in this case.

**Q3. (10 Marks)**

- a. Discuss Intelligent Agents and their types and describe how they interact with environment. CO1

OR

- b. Consider the following weighted graph with six nodes node A as start and node F as goal: CO2

Nodes: A, B, C, D, E, F

Edge Costs:



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- $A \rightarrow B = 2$
- $A \rightarrow C = 3$
- $B \rightarrow D = 4$
- $C \rightarrow D = 1$
- $C \rightarrow E = 5$
- $D \rightarrow F = 2$
- $E \rightarrow F = 2$

| Node | $h(n)$ |
|------|--------|
| A    | 6      |
| B    | 5      |
| C    | 3      |
| D    | 2      |
| E    | 2      |
| F    | 0      |

- i. Apply the A\* algorithm to find the optimal path from A to F.
- ii. Compute the values of  $g(n)$ ,  $h(n)$ , and  $f(n)=g(n)+h(n)$  at each step.
- iii. Identify the final optimal path and total path cost.

Q4. (10 Marks)

- a. i. Discuss what data cleaning is and the need of cleaning the data before applying machine learning.
- ii. Mention any 3 different types of impurities present in the dataset and how to remove them.

CO3

OR

- b. Explain Agent Architecture in detail. Discuss different types of environments with real-life examples. CO1

Q5. (10 Marks)

- a. A retail company wants to predict customer purchase behaviour using the following dataset and perform the following: CO3
  - i. Identify and handle the missing values.
  - ii. Detect and remove outliers in monthly spend using the z-score technique.
  - iii. After cleaning the workout hours column, assume workout hours = [5, 6, 4, 5], apply min-max normalization on workout hours column.
  - iv. Given standard deviation as 1.43, calculate the standardized value for age = 19 by applying standard scalar after removing inconsistency in age column.

| Customer ID | Age | Monthly Spend | Workout Hours | Membership Type |
|-------------|-----|---------------|---------------|-----------------|
| 1           | 21  | 1500          | 5             | Basic           |
| 2           | -5  | 1800          | 6             | Premium         |
| 3           | 20  | 12000         | 4             | Basic           |
| 4           | 18  | 1600          | NULL          | Premium         |
| 5           | 26  | 1550          | 5             | Basic           |

OR

- b. Explain the difference between the following with the help of examples: CO3
  - i. AI, ML, Deep Learning
  - ii. Supervised and Unsupervised Learning



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Note For the question paper setters:

- Question paper should cover all the COs of the course.
- Please specify COs against each question.